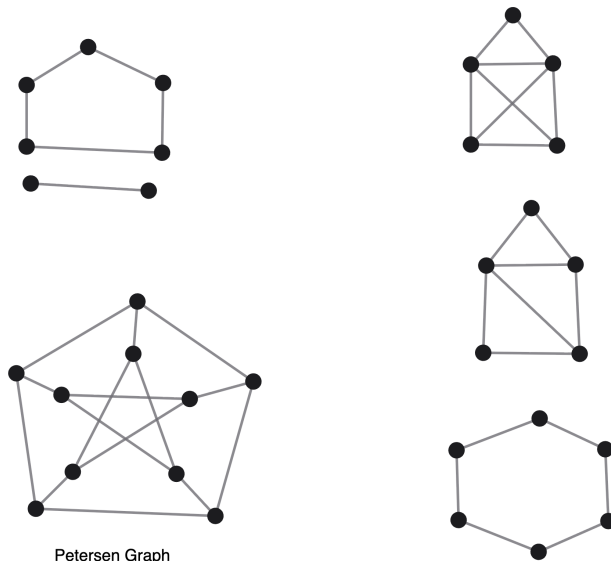


1. Find the chromatic number of the following graphs.



Solution: Left column: 3 and 3. Right column: 4, 3, and 2.

2. Let K_n be the graph on n vertices containing all possible edges (complete graph). What is $\chi(K_n)$?

Solution: $\chi(K_n) = n$.

3. Let C_n be the cycle graph on n vertices. What is $\chi(C_n)$? Prove a formula in general for even and odd n .

Solution: When n is even, we can color the graph red-green-red-green and so on, so $\chi(C_n) = 2$. When n is odd, we can almost do this, but we get a collision at the end, requiring a third color, so $\chi(C_n) = 3$.

4. Let $G = (V, E)$ be a 2-colorable graph.¹ Show that $|E| \leq |V|^2/4$.

Hint: Write $|V| = a + b$ where a and b are the number of red/green vertices. Prove and use the inequality $2ab \leq a^2 + b^2$.

¹These graphs are also called *bipartite*

Solution: First, let us assume the inequality is true, that is to say,

$$2ab \leq a^2 + b^2.$$

Now, every edge must connect a red vertex to a green vertex, so the number of edges is at most ab . Therefore

$$\begin{aligned} |V|^2 &= (a + b)^2 \\ &= a^2 + 2ab + b^2 \\ &\geq 2ab + 2ab \\ &\geq 4|E|. \end{aligned}$$

Finally, we prove the inequality, which we do by noting that

$$a^2 + b^2 - 2ab = (a - b)^2 \geq 0.$$